Human-Computer Interaction – Needfinding Report

Project Topic: Move & Learn: Motor Skills Development App

1. **Team Information**

Name	**Task**	**Participant Type**
Setara Ahmadian	Observation	Children with cerebral palsy/Parents and caregivers/Educators and therapists
Alia Rezae	Online Observation	Children with mild cerebral palsy affecting arm coordination and mild motor coordination difficulties. (Parents)
Mahboba Sakhizada	Observation	Children with cerebral palsy/siblings and parents.
Zulaikha Mohammadi	Observation + Online Observation	Physical therapist and parents.

2. **Individual Observations/ Interviews**

Participants – Observed by Setara Ahmadian
User Profile: Child with severe cerebral palsy (featured in video)
Task Observed: Using a math-based educational therapy game designed for children with CP

Observation Notes:

- Children interacted with the game using limited movements.
- The game interface was designed for slow, simplified gestures (e.g., single-button presses).
- Positive reactions were observed when the child received visual and sound feedback.
- The game combined educational math tasks with physical interaction.

- Developers and therapists co-designed the game based on real user needs.

Post-Observation Summary: (based on statements made by developers and therapists in the video)

- Emphasis on the importance of designing for accessibility and individual abilities.
- Games should be tailored specifically for the needs of children with motor impairments.
- Feedback and motivation are critical in sustaining engagement.
- The game must provide therapeutic benefits without feeling like traditional therapy

Participant 2 – Observed by Setara Ahmadian
User Profile: Group of teenage boys with cerebral palsy and significant access challenges
Task Observed: Free break time in a school setting, independently managing their interactions and environment without adult intervention

Observation Notes:

- Boys displayed a high level of independence and resourcefulness despite their physical and mobility limitations.
- They used assistive devices, such as wheelchairs and communication aids, effectively to interact with each other.
- Group dynamics showed positive social interaction, with humor and camaraderie helping atmosphere.
- Physical space and environment posed significant mobility challenges, yet participants adapted creatively.
- Some participants struggled with fine motor skills, limiting their ability to manipulate certain objects or tools easily.
- Limited environmental accessibility was evident but did not diminish the participants' enthusiasm or agency.
- Participants showed a strong desire for engagement, autonomy, and social inclusion.

Post-Observation Summary:

- Participants expressed a need for improved physical accessibility in and around the school to reduce barriers.

- A greater variety of adaptable recreational tools and games tailored to different physical abilities was desired.

- Emphasis on social opportunities and spaces for interaction without constant adult supervision was important.

- Participants wanted more control over their break time activities with accessible options for independent decision-making.

- Communication technologies and aids should be more seamlessly integrated to support varied individual needs.

Participant 3: Observed and Interviewed by Alia Rezae

User Profile: Mother of Frohar – a 14-year-old girl with mild cerebral palsy affecting arm coordination
Mode: Direct online interview and observation via WhatsApp video call
Location: Sweden (participant) / Bangladesh (interviewer)

Date: 7/27/2025

Background:

Frohar is my niece, and her mother is my sister. They have been living in Sweden for about five years. Since I have seen Frohar grow and heard regularly about her therapy routine, I felt this was a meaningful opportunity to observe and learn more deeply about her motor skill development at home.

Observation Summary:

I observed Frohar using a tablet while sitting on a floor mat with her mother beside her. The app involved dragging objects into a basket. Frohar started off motivated, but when her fingers could not move objects accurately or fast enough, her body language changed. She began moving around, sighing, and asking if she could do something else. Her mother encouraged her with gentle words, guiding her when needed, and praised her even when the action was not completed perfectly.

Notable Observations:

- Frohar performs better when tasks are slower-paced and have audio-visual rewards.

- She gets visibly discouraged by repeated failure or high-precision tasks.
- Her mother is an active supporter during therapy sessions, emotionally and physically.
- Frohar preferred switching to something more entertaining after just 10–12 minutes.

Interview Summary:

The conversation flowed naturally, as we often talk casually, but this time I asked more structured questions.

Q: Do you follow a fixed therapy routine with Frohar?
A: "We try to, but it depends on her energy. She's a teenager now, and sometimes she just wants to read or rest."

Q: What does she find the hardest?
A: "Fast movements and anything that requires her to use both hands in coordination."

Q: What kind of apps does she enjoy using?
A: "She likes colorful ones with fun music or rewards. If the app just gives instructions without any feedback, she loses interest."

Q: What changes would you make to such apps?
A: "I would add more variety. Something new every day so she doesn't feel it is the same routine. Also, the app should adapt. Sometimes it is too easy, sometimes suddenly too hard."

Participant 4: Interview Conducted by Proxy, Summarized by Alia Rezae

User Profile: Mother of Emil, an 11-year-old boy in Sweden with mild motor coordination difficulties
Mode: Indirect interview. I provided questions to my sister, who interviewed in person.
Location: Sweden (participant and proxy) / Bangladesh (interviewer)

Date: 7/27/2025

Background:

I couldn't conduct this interview personally, so I asked my sister in Sweden to help. Her daughter, Frohar, attends the same school as Emil, an 11-year-old boy who also experiences some motor coordination challenges. Emil's mother and my sister are acquainted through school events. With consent, my sister visited their home and asked the interview questions on my behalf.

Observation Summary (Based on My Sister's Report):

My sister observed Emil during a short session where he used an app that involved tracing shapes and dragging elements into place. Emil appeared excited at first but quickly became restless. When he had difficulty tracing the shapes correctly, he frowned and asked for help.

His mother offered calm support, but after about 10 minutes, Emil wanted to switch to YouTube.

Reported Observations:

- Emil enjoys the beginning of tasks but gets discouraged when precision is required.
- He is highly motivated by reward-based feedback like stars and sounds.
- He disengages quickly if tasks repeat or if he encounters multiple failed attempts.
- His mother was patient and attentive but noted that keeping him focused is challenging.

Interview Summary (Via My Sister):

My sister sent me voice notes summarizing the mother's responses.

Q: How often do you practice these exercises at home?
A: "A few times a week, usually after school. I try not to make it feel like a task. He cooperates more when it is playful."

Q: What frustrates Emil most in these apps?
A: "Tasks that need very exact finger control. He gets tired and gives up if he does not succeed quickly."

Q: What works best for him?
A: "Visual rewards. If the app praises him, like with sounds or stars, he becomes more confident. Also, he likes when he can take his time without being rushed."

Q: What would you improve?
A: "I would like the app to slowly adjust to his level. Sometimes the games become harder too quickly. Also, I wish there were more different games in one app. He gets bored easily."

Q: Are these apps helpful overall?
A: "Definitely. When they are well designed, he learns while playing. But they have to stay engaging. If not, he switches to entertainment apps."

Participants 5: Observed by Mahboba Sakhizada
User Profile: Child with cerebral palsy (featured in video)
Task Observed: Playing an instrumental based video game designed for children with cerebral palsy.

Observation Notes:

- Jillian Peters playing the game using real instruments.

- The game interface was designed for more engaging activities for children and also their siblings to engage with them.

- She is very happy hearing the sound of instruments, she enjoys clacking things together.

- The game combined more physical activities with sounds, which helps them as a therapy.

Post-Observation Summary: *(based on statements made by the user and developers in the video)*

- Emphasis on the importance of engaging activities that can be played at home, with parents and siblings.
- Repetitive movements are helpful for children with CP.
- Constraint therapy can also be used to help children learn movements that they aren't familiar with.
- Traditional therapy can be helpful if made in an engaging and fun way.
- The involvement of parents and siblings in gameplay can create a supportive, social environment, enhancing emotional bonding and reducing feelings of isolation.
- The game interface might offer adjustable difficulty or customizable input methods to accommodate the wide range of motor abilities found in children with CP.
- The physical interaction with real instruments may translate better to real-world use and improve hand-eye coordination.

Participant 6 – Observed by Mahboba Sakhizada
User Profile: Child with moderate cerebral palsy (featured in video)
Task Observed: Playing a video game using an inclusive gaming controller specially designed for the child.

Observation Notes:

- Jerome played the game using a big game controller specially designed for him in the university of Melbourne.

- The game controller was designed by university students for eight months, they did close observations on how to track the movements of Jerome.

- Jerome is very happy to be able to play games with a big controller.

- The controller registers the electrical activities in his skin and outputs it as computer keys.
- Playing video games before the big controller was impossible for children with CP.

Post-Observation Summary: *(based on statements made by the parents and developers in the video)*

- His facial expressions, excitement, and willingness to engage showed just how powerful accessible design can be.

- The technology used for this controller can be used for other children as well and customized according to their needs.

- The controller uses many sensors that can be helpful in different areas.

- The game controller is big enough which makes it easier for Jerome to access different buttons.
- Because of the video game it is played on a big screen where it makes it more interesting for the user and more engaging.
- The big screen is easier to track movements of the video played on the screen and also receive necessary feedback.
- In this case the feedback is both visual and audio feedback.

Participant 7: Observed by Zulaikha Mohammadi (featured in Video)

User Profile:
Physical therapist who works with children (ages 6–10) diagnosed with mild to moderate cerebral palsy. The therapist regularly uses a VR-based motor skill therapy system in clinical sessions.

Task Observed:
Using a VR game to help an 8-year-old child with mild cerebral palsy improve upper limb coordination during a therapy session.

Observation Notes:

- The child wore a lightweight VR headset and interacted with a virtual environment using hand and arm movements.

- The game environment encouraged reaching, catching, and pushing objects targeting fine and gross motor skill improvement.

- The child showed high levels of engagement, smiling and laughing during the session.

- The game responded with visual and audio feedback (e.g., a “ding” sound and animation when the child succeeded).

- The therapist adjusted difficulty levels manually depending on how easily the child completed each task.

- The system recorded performance data like number of successful movements and time spent on each task.

- Despite mild coordination difficulty in the child's right hand, the VR interface was generally accessible and supportive.

- Therapists used verbal prompts alongside the game to encourage specific movements.

- No physical controls were used; the interface was gesture-based (via motion sensors).

Post-Observation Summary:

- Participants expressed that the VR game is much more engaging than traditional therapy methods, leading to increased motivation and participation in motor skill exercises.

- Participants noted challenges with the setup process, highlighting that the headset can be a barrier for children with attention difficulties or visual impairments.

- There was a desire for better progress tracking and adaptive task suggestions to tailor therapy more effectively.

- Participants emphasized the need for a mobile version or lighter hardware that could be used conveniently at home.

- Personalization based on individual motor abilities, such as differences between left and right hand strength, was identified as an important feature.

- Participants highlighted the value of real-time feedback systems for caregivers and remote therapists to monitor and support the child's progress.

- Caregivers expressed a need for the system to be simple to use, considering that not all parents are tech-savvy, so they can safely guide their children and track progress.

Participant 8: Interview Conducted by Zulaikha Mohammadi
Platform: WhatsApp (voice messages and follow-up texts)
Date: 26/07/2025
Duration: Approximately 20 minutes

User Profile

The participant is the mother of Ibrahim, a 19-year-old boy living in Germany with mild cerebral palsy, which mainly affects his left arm and leg, leading to coordination difficulties.

The family relocated to Germany three years ago and receives government support, including regular consultations with physical therapists, access to assistive devices, and medical care. At home, the mother takes charge of daily therapy sessions and uses digital apps to guide and support their routine.

Introduction

Zulaikha began by greeting the participant and explaining the goal of the interview: to better understand home-based caregiving for individuals with motor disabilities and explore the use of digital tools, especially apps, in improving therapy engagement and caregiver support.

Interview Highlights

Zulaikha: Could you describe your daily routine with Ibrahim, especially regarding therapy activities?
Mother:
"Sure. Every morning, we spend about 20 to 30 minutes doing therapy. I use an app called *TherapyTrack* that gives us a list of customized exercises for Ibrahim. We do stretches, tapping games, and some hand-eye coordination tasks. The app uses animations and sound prompts, which really help hold his attention."

Zulaikha: That sounds helpful. Which apps are you currently using?
Mother:
"We mainly use *TherapyTrack* for physical exercises, and sometimes *MotriPlay*, which is a fun app with mini-games designed for kids and teens with motor difficulties. Ibrahim prefers MotriPlay because it feels more like a game. It gives stars, claps, and cheerful sounds when he completes a task that really boosts his motivation."

Zulaikha: How have these apps changed your experience as a caregiver?
Mother:
"A lot, actually. Before using apps, I used to rely on printed exercises and YouTube videos, and I had to guess what worked and what did not. Now, the apps track his progress and remind me what to do next. On days when I am tired or unsure what activity to plan, the app gives clear guidance. It also helps me stay organized."

Zulaikha: Does Ibrahim enjoy using the apps directly himself?
Mother:
"Yes, definitely. He likes that he can tap on things, drag shapes, or follow along with animated characters. It makes him feel more in control and less like he is being 'forced' to do therapy. He even reminds me sometimes, saying, 'Let's do the robot game today!' So now it is not just therapy, it is something he looks forward to."

Zulaikha: What features do you find most helpful in these apps?
Mother:
"The best part is the personalization. Both apps adjust the difficulty based on how he performs. They give encouragement like stars, claps, or even badges. There is also a parent

dashboard where I can see his weekly progress. One of the apps allows me to send reports directly to his physical therapist that saves me from explaining everything during appointments."

Zulaikha: Do you still face any challenges, even with the apps?
Mother:
"Yes, of course. Apps are helpful, but they are not perfect. Sometimes Ibrahim still gets frustrated if a task is too hard. And I still need to supervise everything. He can't use the app fully independently yet. Also, some of the better apps require subscriptions, and we have to be careful with our budget."

Zulaikha: Is there anything else that could make digital therapy tools even better?
Mother:
"I would love to see more apps made for teenagers. Most of what is out there feels too childish or too advanced. Also, having offline access would help when the internet is slow. And maybe a way to connect with other parents using the same app like a small support group built into it."

Zulaikha: Does the German healthcare system support the use of such apps?
Mother:
"Not directly. They do not prescribe the apps, but some therapists recommend them unofficially. We bought them on our own after one of Ibrahim's therapists showed us a demo. I think these tools should be more widely promoted. They make a big difference at home, especially for families like ours."

3. Consolidated User Needs
After reviewing all observations and interviews, the following user needs were identified:

1. Simplified, adaptive interfaces are essential; children benefit from large buttons, minimal screens, and gesture-based inputs that accommodate various motor abilities.

2. Positive feedback systems like stars, music, clapping, and animations significantly increase motivation and reduce frustration.

3. Customizable difficulty and pacing are crucial, as some users get discouraged by tasks that are too fast, repetitive, or precise.

4. Parental involvement and controls are necessary. Caregivers want the ability to supervise, adjust content, and support their child's use of the app.

5. Progress tracking dashboards help parents and therapists monitor improvements, therapy time, and patterns of disengagement.

6. Therapeutic engagement must feel playful, not clinical games should feel entertaining while achieving motor development goals.

7. Offline functionality and low-tech compatibility are important for families with limited internet access or older devices.

8. Physical and sensory accommodations, like contrast settings, simplified visuals, and support for one-handed use, are needed to prevent overstimulation.

9. Peer and family integration features (e.g., multiplayer mode or joint activities) promote social bonding and shared play experiences.

10. Content variety and regular updates keep children engaged longer and prevent boredom during therapy routines.

11. Real-world interaction features, such as using real instruments or physical toys, help bridge in-game movement to real-life skills.

12. Therapists and educators need professional tools, such as customizable therapy goals, session logs, and remote monitoring access.

13. Age-appropriate design matters: apps should consider content appeal for different developmental stages, especially for teens who outgrow childish interfaces.

14. Financial accessibility and app affordability were recurring concerns for families relying on free or low-cost resources.

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